

Generalized Noether Theory in Twin Spaces: Symmetry, Arithmetic Deformations, and Conserved Quantities

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Abstract

We generalize Noether’s theorem to *twin spaces*: structures in which the arithmetic of the configuration space is deformed by a group action. Given a set E with a reference operation α and a group G acting by bijections, each $g \in G$ induces a transported arithmetic $x \diamond_g y = g^{-1}(g(x) \alpha g(y))$. We develop the Lagrangian formalism in each arithmetic $\diamond_g$, prove the Euler–Lagrange equations by transport of structure (Theorem 10), and establish a *parametrized Noether theorem*: every continuous symmetry of a g -Lagrangian yields a conserved quantity C_g expressed in the arithmetic $\diamond_g$ (Theorem 13). The family $\{C_g\}_{g \in G}$ transforms covariantly under G , revealing a form of *arithmetical relativity*: conserved quantities depend on the arithmetic used to describe the system. We prove that a symmetry hidden in one arithmetic may become manifest in another (Theorem 15). Applications to multiplicative mechanics (dilation symmetry $\rightarrow$ relative momentum $m\dot{x}/x$), logarithmic oscillators (Gompertz dynamics), and scale-invariant cosmology illustrate the framework. The results connect to the (G, H) -equivariant theory of twin information theory and to the twin variational calculus of twin general relativity.

Keywords: Noether theorem, twin spaces, deformed arithmetic, conservation laws, arithmetical relativity, Lagrangian mechanics

1 Introduction

Noether’s theorem [1] is formulated on the configuration space $\mathbb{R}^n$ with its standard arithmetic $(+, \cdot)$. Velocities, momenta, and energies all use the additive structure of $\mathbb{R}$.

However, many physical systems are more naturally described by non-additive structures: pop-

ulation growth (multiplicative rates $\dot{x}/x$), financial models (compounded returns), perception (Weber–Fechner logarithmic law), and cosmological expansion (Hubble parameter $H = \dot{a}/a$). In these regimes, the “natural” primitive operation is not addition but a deformed variant.

The twin spaces framework [4, 5] provides a controlled deformation: given a bijection $g : E \rightarrow E$, the transported operation $x \diamond_g y = g^{-1}(g(x) + g(y))$ is isomorphic to addition but has different algebraic expressions. Each g induces its own differential calculus, Lagrangian theory, and conservation laws.

Our main result: a Noether theorem in each arithmetic $\diamond_g$, producing a G -indexed family of conserved quantities. The classical theorem is the special case $g = \text{id}$.

2 Twin Spaces and Deformed Arithmetics

Definition 1 (Twin Space). A **twin space** is a triple (E, α, G) where E is a set, $\alpha : E \times E \rightarrow E$ is a binary operation (the reference), and G acts on E by bijections. For each $g \in G$, the **g -deformed operation** is:

$$x \diamond_g y := g^{-1}(g(x) \alpha g(y)) \quad (1)$$

Proposition 2 (Transport of Structure). *The map $\varphi_g : (E, \diamond_g) \rightarrow (E, \alpha)$, $\varphi_g(x) = g(x)$, is an isomorphism of magmas. If α is associative (resp. commutative, has identity e), then $\diamond_g$ is associative (resp. commutative, has identity $g^{-1}(e)$).*

Proof. $\varphi_g(x \diamond_g y) = \varphi_g(g^{-1}(g(x) \alpha g(y))) = g(x) \alpha g(y) = \varphi_g(x) \alpha \varphi_g(y)$. Bijectivity of g gives the isomorphism. Properties transfer through isomorphisms. $\square$

Definition 3 (Kernel of the Arithmetic Action). The **arithmetic kernel** $\ker \pi = \{g \in G : \diamond_g = \alpha\}$ is the subgroup preserving the reference arithmetic. The quotient $G/\ker \pi$ indexes the distinct arithmetics.

Example 4 (Exponential Hierarchy). $E = \mathbb{R}_{>0}$, $\alpha = +$, $G = \langle \exp \rangle \cong \mathbb{Z}$. Let $g_n = \exp^{(n)}$ (n -fold iterated exponential). Then:

- $n = 0$: $\diamond_0 = +$ (ordinary addition).
- $n = 1$: $x \diamond_1 y = \log(e^x + e^y)$ (log-sum-exp).
- $n = -1$: $x \diamond_{-1} y = \exp(\log x + \log y) = x \cdot y$ (ordinary multiplication, since $g_{-1} = \log$, $g_{-1}^{-1} = \exp$).

The hierarchy interpolates between additive, logarithmic, and multiplicative regimes.

3 Lagrangian Formalism in Deformed Arithmetic

Fix (E, α, G) and $g \in G$. Let $x : [t_1, t_2] \rightarrow E$ be a smooth curve.

Definition 5 (g -Velocity). The g -velocity of x is the velocity of the transported curve $u = g \circ x$,

$$v_g(t) := \frac{d}{dt} g(x(t)) = dg_{x(t)}[\dot{x}(t)], \quad (2)$$

a tangent vector at $u(t) = g(x(t))$. It is the rate of change of x measured in the reference arithmetic.

Remark 6. The outer g^{-1} does *not* appear: $g^{-1}(\frac{d}{dt}g(x))$ is not a well-defined derivative (the difference quotient $g^{-1}(\dot{u}h)/h$ diverges unless $g^{-1}(0) = 0$). The well-defined object is the transported tangent vector $\dot{u} = dg_x[\dot{x}]$.

Example 7 (Multiplicative Velocity). For $g = \log$, $x(t) > 0$: $v_{\log}(t) = \frac{d}{dt} \log x(t) = \dot{x}/x$, the relative growth rate. In the variable $y = \log x$ this is simply $\dot{y}$.

Definition 8 (g -Lagrangian and g -Action). Given a classical Lagrangian $L : TE \rightarrow \mathbb{R}$, the g -**Lagrangian** is $L_g(x, v) := L(g(x), v)$ (the second slot already being a velocity in the reference arithmetic), so that $L_g(x, v_g) = L(u, \dot{u})$ with $u = g(x)$. The g -**action** is:

$$J_g[x] := \int_{t_1}^{t_2} L_g(x(t), v_g(t)) dt \quad (3)$$

Remark 9 (Reading L_g). The two slots of L_g play asymmetric roles: the position is deformed ($x \mapsto g(x)$), whereas the velocity slot already carries a reference-arithmetic velocity $v_g = \frac{d}{dt}g(x)$.

Equivalently, L_g is the pullback of L along the point transformation $u = g(x)$, lifted to velocities by dg_x . This is precisely why the g -dynamics is the classical dynamics read in the chart $u = g(x)$, as Theorem 10 makes exact.

Theorem 10 (Euler–Lagrange in Deformed Arithmetic). A curve x with fixed endpoints is a critical point of J_g iff its transport $u = g(x)$ satisfies the classical Euler–Lagrange equation:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{u}}(u, \dot{u}) \right) - \frac{\partial L}{\partial u}(u, \dot{u}) = 0, \quad u = g(x), \quad \dot{u} = v_g. \quad (4)$$

Proof. Let $u(t) = g(x(t))$. Then $J_g[x] = \int L(u, \dot{u}) dt$ (by definition of L_g and v_g). A variation x_ε with fixed endpoints lifts to $u_\varepsilon = g(x_\varepsilon)$ with fixed endpoints. The first variation $\delta J_g = \int \langle \frac{\partial L}{\partial u} - \frac{d}{dt} \frac{\partial L}{\partial \dot{u}}, \delta u \rangle dt = 0$ is the classical Euler–Lagrange equation in u -coordinates. Since $u = g(x)$ is a diffeomorphism and the variation has fixed endpoints, criticality of J_g is equivalent to the classical Euler–Lagrange equation (4) for u . $\square$

Remark 11 (Covariance). The dynamical equations are covariant under arithmetic deformation: changing g changes the coordinate representation but not the geometric content. This is the Lagrangian version of the “graph invariance” principle of the (G, H) -equivariant framework [7].

4 The Generalized Noether Theorem

Definition 12 (g -Symmetry). A one-parameter family (τ_ε) on E is a g -**symmetry** of J_g if $J_g[\tau_\varepsilon \circ x] = J_g[x] + o(\varepsilon)$ for all smooth x . Its infinitesimal generator is $X(x) = \frac{d}{d\varepsilon}|_{\varepsilon=0} \tau_\varepsilon(x)$.

Theorem 13 (Noether Theorem in Twin Spaces). Let (τ_ε) be a g -symmetry of J_g with generator X . For every solution $x(t)$ of (4), the quantity

$$C_g(x, v_g) := \left\langle \frac{\partial L_g}{\partial v_g}(x, v_g), dg_x[X(x)] \right\rangle = \left\langle \frac{\partial L}{\partial \dot{u}}, Y(u) \right\rangle \quad (5)$$

is conserved: $\frac{d}{dt} C_g(x(t), v_g(t)) = 0$, where $u = g(x)$ and $Y(u) = dg_x[X(x)]$ is the transported generator. (The generator X must be carried to the reference space by dg_x ; using X in place of $dg_x[X]$ gives a quantity that is generally not conserved.)

Proof. Step 1 (Lift). Let $u = g(x)$. The transformation $\sigma_\varepsilon(u) = g(\tau_\varepsilon(g^{-1}(u)))$ is a one-parameter family on the u -space with generator $Y(u) = dg \cdot X(g^{-1}(u))$.

Step 2 (Classical Noether). The g -symmetry condition lifts to invariance of $J[u] = \int L(u, \dot{u}) dt$ under σ_ε . The classical Noether theorem [2] gives: $C(u, \dot{u}) = \langle \frac{\partial L}{\partial \dot{u}}, Y(u) \rangle$ is conserved.

Step 3 (Transport back). Since $L_g(x, v) = L(g(x), v)$ we have $\frac{\partial L_g}{\partial v_g} = \frac{\partial L}{\partial \dot{u}}$ at $u = g(x)$, and $Y(u) = dg_x(X(x))$. Hence $C(u, \dot{u}) = \langle \frac{\partial L}{\partial \dot{u}}, Y(u) \rangle = \langle \frac{\partial L_g}{\partial v_g}, dg_x[X(x)] \rangle = C_g(x, v_g)$. $\square$

Corollary 14 (Family of Conservation Laws). *For each $g \in G$, there exists a conserved quantity C_g . The family $\{C_g\}_{g \in G}$ transforms covariantly: $C_t = C_s \circ f_{s,t}$ where $f_{s,t} = g_s \circ g_t^{-1}$.*

Theorem 15 (Hidden Symmetries). *A symmetry (τ_ε) that is broken for J_{g_1} may be exact for J_{g_2} (with $g_1, g_2 \in G$, $g_1 \neq g_2$). The corresponding conserved quantity C_{g_2} , when expressed in the g_1 -arithmetic, appears as a **hidden invariant** — a function conserved along trajectories but not derivable from a manifest symmetry in $\diamond_{g_1}$.*

Proof. The invariance condition $J_g[\tau_\varepsilon \circ x] = J_g[x]$ depends on g through the g -Lagrangian L_g . Changing g to $g' = h \circ g$ transforms L_g to $L_{g'}$, which may have different symmetries. Specifically: $L_{g'}(x, v) = L(h(g(x)), dh \cdot dg[v])$, and the extra factor dh may restore or break invariance under τ_ε . $\square$

Example 16 (A symmetry hidden by the choice of arithmetic). Fix $L(u, \dot{u}) = \frac{m}{2}\dot{u}^2$ and the dilation $\tau_\varepsilon(x) = e^\varepsilon x$. In the *standard* arithmetic ($g = \text{id}$) the action $J_{\text{id}}[x] = \int \frac{m}{2}\dot{x}^2 dt$ is *not* dilation-invariant: $\dot{x} \mapsto e^\varepsilon \dot{x}$ rescales the integrand by $e^{2\varepsilon}$, so the symmetry is broken. In the *multiplicative* arithmetic ($g = \log$) the same dilation acts on $v_g = \dot{x}/x$ by $\dot{x}/x \mapsto (e^\varepsilon \dot{x})/(e^\varepsilon x) = \dot{x}/x$, leaving $J_{\log}[x] = \int \frac{m}{2}(\dot{x}/x)^2 dt$ invariant. Thus one and the same symmetry is broken for J_{id} yet exact for $J_{\log}$, exactly as in Theorem 15. Its twin Noether charge $C_{\log} = m\dot{x}/x$ is conserved along the $J_{\log}$ -trajectories $x(t) = x_0 e^{at}$; read back in the standard arithmetic it is the non-obvious invariant $m\dot{x}/x$ — conserved along these trajectories but not the charge of any manifest symmetry of J_{id} , i.e. a hidden invariant.

5 Applications

5.1 Multiplicative Free Particle

$E = \mathbb{R}_{>0}$, $g = \log$ (so $u = \log x$ and $v_g = \dot{x}/x$). In the variable $y = \log x$: $L_\times = \frac{m}{2}\dot{y}^2 = \frac{m}{2}(\dot{x}/x)^2$.

The Euler–Lagrange equation $\ddot{y} = 0$ gives $x(t) = x_0 e^{a(t-t_0)}$ (exponential trajectory — constant relative growth rate).

Proposition 17 (Dilation $\rightarrow$ Relative Momentum). *The dilation $\tau_\varepsilon(x) = e^\varepsilon x$ (generator $X(x) = x$) is a symmetry of $L_\times$. The Noether invariant is $C_\times = m\dot{x}/x$ — the **relative (logarithmic) momentum** $m\dot{x}$.*

Proof. In y -coordinates τ_ε becomes $y \mapsto y + \varepsilon$ (translation); since $L_\times = \frac{m}{2}\dot{y}^2$ is independent of y it is invariant. The transported generator is $Y = dg_x[X] = g'(x)x = (1/x)x = 1$, so the Noether charge is $C = \frac{\partial L_\times}{\partial \dot{y}} Y = m\dot{y} = m\dot{x}/x$. This is conserved because $\ddot{y} = 0 \Rightarrow \dot{y}$ is constant; note that $m\dot{x}$ itself is *not* conserved, since $\dot{x} = x\dot{y}$ grows exponentially. $\square$

Remark 18 (Arithmetical relativity in action). In the standard arithmetic ($g = \text{id}$): translation symmetry $x \mapsto x + \varepsilon$ gives the additive momentum $p = m\dot{x}$. In the multiplicative arithmetic ($g = \log$): the multiplicative translation (dilation) $x \mapsto e^\varepsilon x$ gives the $\diamond_{\log}$ -momentum $m\dot{x}/x = m\dot{y}$. The conservation structure (translation $\rightarrow$ momentum) is the same, but the conserved invariant is the one appropriate to each arithmetic. This is arithmetical relativity: which quantity is conserved depends on the arithmetic used to describe the system.

5.2 Logarithmic Harmonic Oscillator

$L_{\log} = \frac{m}{2}(\dot{x}/x)^2 - \frac{k}{2}(\log(x/x_0))^2 = \frac{m}{2}\dot{y}^2 - \frac{k}{2}y^2$ with $y = \log(x/x_0)$.

Solution: $x(t) = x_0 \exp(A \cos \omega t + B \sin \omega t)$ with $\omega = \sqrt{k/m}$. This is a **multiplicative oscillation** around x_0 : the ratio x/x_0 oscillates on a logarithmic scale.

Proposition 19 (Logarithmic Energy). *Time-translation invariance gives the conserved **logarithmic energy**: $E_{\log} = \frac{m}{2}(\dot{x}/x)^2 + \frac{k}{2}(\log(x/x_0))^2$.*

Such oscillators model Gompertz growth, Weber–Fechner perception, and critical fluctuations on logarithmic scales.

5.3 Scale-Invariant Cosmology

The Hubble parameter $H = \dot{a}/a$ is a multiplicative velocity. The Friedmann Lagrangian $L_{\text{cosmo}} = \frac{M_{\text{Pl}}^2}{2}H^2 - V(a)$ is naturally a g -Lagrangian with $g = \log$.

If $V(a) = \Lambda$ (cosmological constant): L is invariant under $a \mapsto \lambda a$ (multiplicative dilation). By Theorem 13: there exists a conserved quantity associated with this scale symmetry, constraining $H(t)$ and $a(t)$.

Remark 20 (Connection to Vol. 17). This is the 1D orbital reduction of FLRW cosmology (Vol. 17, [7]): the Friedmann equations are the Euler–Lagrange equations of the g -action on the 1D orbit space, and the scale-invariance conservation law is the twin Noether charge C_g with $g = \log$.

6 Connection to the (G, H) -Equivariant Framework

The Noether theorem in twin spaces connects to two other parts of the ROOTS-INSIGHTS program:

1. **Twin Information Theory** (Vol. 16, [6]): the optimal group selection problem asks which $g \in G$ minimizes the coding cost. The twin Noether theorem asks which g reveals the most symmetries. Both are optimization problems over the group G , one information-theoretic, the other variational.
2. **Twin General Relativity** (Vol. 17, [7]): the twin Einstein equations $\tilde{G}_{\mu\nu} + \tilde{J}_{\mu\nu} = 8\pi\tilde{T}_{\mu\nu}$ are the field-theoretic version of the g -Euler–Lagrange equations (4), and the twin Noether charge of §4 is the Lagrangian version of the tensorial twin Noether current $\tilde{J}_\xi^\mu = \tilde{T}^{\mu\nu}\tilde{\xi}_\nu|J|$.
3. **The Splitting Theorem** ($\mathfrak{g} = \mathfrak{ker} \oplus \mathfrak{def}$): the symmetries producing conservation (Noether) live in $\mathfrak{ker}(g)$; the deformations producing dynamics live in $\mathfrak{def}(g)$. The twin Noether theorem is the variational expression of this splitting.

7 Conclusion

We have shown that Noether’s theorem extends naturally to twin spaces: every continuous symmetry of a g -Lagrangian yields a conserved quantity C_g in the arithmetic $\diamond_g$. The family $\{C_g\}_{g \in G}$ reveals arithmetical relativity — conservation laws

depend on the arithmetic — and hidden symmetries that are manifest in one arithmetic but broken in another.

Open problems: (1) Quantization in twin spaces: Hilbert spaces with deformed inner products, spectra under arithmetic deformation. (2) Selection of optimal arithmetic: variational principle for the arithmetic itself, connecting to the optimal group problem of [6]. (3) Stochastic twin calculus: Itô/Stratonovich formulations in deformed arithmetics for multiplicative noise models.

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